

Alejandro Sánchez Yalí

Senior Full Stack Engineer



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SUMMARY

Senior Full Stack Engineer with 8+ years building and maintaining scalable web applications in Python and TypeScript. Three years on a distributed team across Canada, the US and Latin America building frontend and backend and reviewing code, followed by two years as an independent consultant designing systems for clients and building three products end to end. M.Sc. in Mathematics. Strong bias toward clean architecture, testing, and decisions that survive their second year.

TECHNICAL SKILLS

- **Languages:** Python (8+ years), TypeScript (6+ years), JavaScript (6+ years), SQL, Rust (personal projects)
- **Backend:** Django, Django REST Framework, FastAPI, Celery, Flask, Node.js, Express, NestJS, RESTful APIs, GraphQL, microservices
- **Frontend:** React.js (7+ years), Next.js (6+ years), Tailwind CSS, Shadcn UI, Redux, Zustand, Zod, SASS
- **Databases:** PostgreSQL, MongoDB, MySQL, Redis, Firebase
- **Cloud & DevOps:** AWS, GCP, Docker, GitHub Actions, CI/CD pipelines
- **Engineering:** Clean architecture, DDD, SOLID, DRY, TDD, Vitest, Jest, React Testing Library, code review, mentoring

EXPERIENCE

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| Full Stack Engineer
<i>Monadical</i> | Apr 2021 – Aug 2024
<i>Remote, Canada/US</i> |
| <ul style="list-style-type: none">– Participated in system design and architectural decision-making for new features and services, evaluating trade-offs across performance, scalability and maintainability– Built and maintained full stack applications in Python (Django, FastAPI) and TypeScript (React.js, Next.js), with RESTful and GraphQL APIs– Developed both frontend and backend, including reusable components and tests (Vitest, Jest, TDD)– Designed ETL pipelines and custom Django ORM extensions; managed deployments on AWS and GCP with Docker and CI/CD– Did code reviews on a distributed team in three countries | |
| Independent Software & AI Consultant
<i>Freelance</i> | Aug 2024 – Present
<i>Remote</i> |
| <ul style="list-style-type: none">– Designed and delivered three products end to end, owning architecture, backend, frontend and deploy for each– Designed the database architecture for a client's workflow automation platform in an NX multi-repo, and served as code reviewer for their team– Ran technical design sessions with clients, translating business requirements into system architectures– Refactored legacy codebases and implemented practices for scalability and security | |
| AI Specialist
<i>Lapzo</i> | Aug 2025 – Sep 2026
<i>Remote, Mexico</i> |
| <ul style="list-style-type: none">– Made the key architectural decisions for an educational platform, designing the integration between services and the real-time communication layer– Implemented real-time features with Next.js, NestJS and Firebase Realtime Database– Built the content-generation pipelines with Celery workers running long jobs in the background | |
| Full Stack Engineer
<i>BCFort</i> | Aug 2018 – Oct 2020
<i>Medellín, Colombia</i> |
| <ul style="list-style-type: none">– Designed system architecture for blockchain and analytics platforms: data models, service boundaries and smart contract patterns– Built platforms with Python (Django), Node.js, React and Web3.js; managed infrastructure on AWS, GCP and IBM Cloud | |

- Mentored junior engineers, supporting their technical growth and code quality

Professor of Mathematics & Physics
Universidad de Antioquia

Jan 2010 – Oct 2021
Medellín, Colombia

- Taught mathematics and physics; in the last year and a half also machine learning and data science with Python and the scientific stack
- Developed curriculum for programming, mathematics and ML, and mentored student research projects

FEATURED PROJECTS

- **VitaStock – Surgical Supply Chain for Clinics**  – *Python, FastAPI, Next.js, TypeScript, PostgreSQL, Docker, Railway*
 - Architect and lead engineer of a platform commissioned by a private surgical clinic in Colombia – 12 modules and 5 user roles designed to replace spreadsheet-based inventory and procurement
 - Surgery scheduling with supply baskets and real-time consumption tracking, lot-level traceability with expiry alerts, and true per-patient cost through a cardex
- **Plixiq – AI Customer Support on WhatsApp**  – *Python, FastAPI, LiteLLM, PostgreSQL, Next.js, Railway*
 - Took the product from an idea to a deployed system as architect and lead engineer: discovery, architecture, backend, frontend and deploy
 - About 40,000 lines to a near-complete MVP in 4 months part-time, running at \$60–80/month because cost scales with LLM usage rather than infrastructure
- **Kipux – Collaborative Finance App in Rust**  – *Rust, Axum, sqlx, PostgreSQL, HTMX, SSE, Shuttle, MIT (work in progress)*
 - Personal project to learn Rust by shipping a real app: multi-tenant projects with invites and roles, budgets, real-time updates over Server-Sent Events, and a PWA with an offline transaction queue
 - Receipt-by-photo parsing with Claude vision through a strict JSON schema; integration tests on isolated migrated databases and CI on GitHub Actions
- **Digital Human – Interactive AI Avatar**  – *React, Three.js, React Three Fiber, OpenAI GPT, Whisper, ElevenLabs*
 - Open-source conversational avatar with 464 stars and 110 forks: real-time speech-to-text, generation, voice synthesis and lip synchronisation
 - The engineering problem was latency budgeting across four services, since a human notices any pause in a conversation

OPEN SOURCE

- **talking-avatar-with-ai**  – 464 stars, 110 forks. Open-source conversational AI avatar: OpenAI GPT, Whisper, ElevenLabs, Rhubarb lip-sync, Three.js
- **Monadical-SAS/Morpheus**  – 19 pull requests. Open-source AI image generation platform: FastAPI, React, Stable Diffusion
- 493 GitHub stars across own open-source projects – top 4.8% of ranked GitHub profiles (GitRanks)

TALKS & WRITING

- **Scaling Learning Models: Parallelism Strategies in JAX with Flax**  – PyCon Colombia, 2024

EDUCATION

M.Sc. in Mathematics
Universidad de Antioquia

Aug 2010 - Sep 2013
Medellín, Colombia

B.Sc. in Teaching Mathematics and Physics
Universidad de Antioquia

Jan 2004 - Oct 2009
Medellín, Colombia